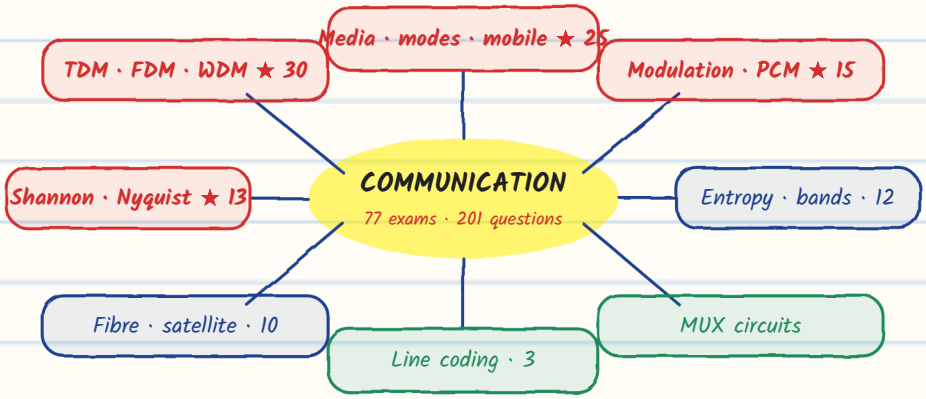
Communication Theory

every communication question asked in BCS · NTRCA · BPSC · bank · BTRC · BUET-set papers, solved in 29 sheets



↑ the number is how many different exams asked that topic (red = most asked)

➜ How to read a question card

Q The question, as printed in the paper (typos fixed).
✎ **Asked in:** the exams it was taken from, e.g. 45th BCS CSE (2024) · BPSC AP 2021

No dated exam → the card names its source (suggestion sheet, bank IT viva). Then: answer → green = worked result → trap.

Most asked

① TDM vs FDM (7 exams)
② $C = B \log_2(1 + S/N)$
③ Nyquist rate of $\sin(2100\pi t)$
④ PCM: blocks, pros/cons
⑤ FDM + guard bands
⑥ TDM frame numericals
⑦ AM / FM bandwidth

➜ Syllabus coverage

"Spectral analysis: Fourier series, sampling function, power spectrum ...; Information theory: entropy, information rate, Shannon's theorem, channel capacity; Analog communication system: different modulations ...; Digital modulation: different standard modulation schemes; Pulse and digital signals: PAM, PCM, DM, ADM; Multiplexing: TDM, FDM, multiple-access network: TDMA, FDMA, CDMA".

Legend yellow = key term · red circle = must-memorise · **purple box** = memory trick · **red box** = exam trap · **green** = worked answer

➜ Contents

✓ Multiplexing: TDM · FDM · WDM ... p. 2–5
✓ MUX / DEMUX circuits ..... p. 6–7
✓ Sampling, Shannon, Nyquist ... p. 7–9
✓ Bit rate, baud, dB ..... p. 10
✓ AM, FM, digital modulation ... p. 11–14
✓ PCM, DM, line coding ..... p. 14–16
✓ Entropy, information rate ... p. 16–17
✓ Bands, channels, modes ..... p. 18–20
✓ Media & optical fibre ..... p. 21–23
✓ Satellite, mmWave ..... p. 23–24
✓ Mobile: GSM, LTE, CDMA, 5G ... p. 25–26
✓ Also filed here ..... p. 27
✓ Night-before sheet ..... p. 28
✓ Practice MCQs + written ..... p. 29

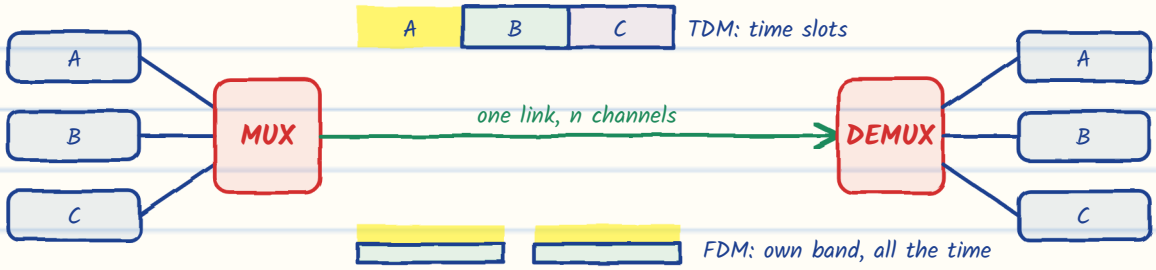
1 TDM vs FDM: the most-asked question

30 exams ask multiplexing

Q1 What is multiplexing in data communication? Write the differences between TDM and FDM.

✎ **Asked in:** 45th BCS CSE (2024) · BPSC AP 2021 · 17th NTRCA CSE (2023) · 17th NTRCA ICT (2023) · 36th BCS (2016)

Definition *Multiplexing* sends several signals over **one shared link** at the same time: a MUX combines n inputs onto one line, a DEMUX separates them at the far end. It is used because one link's capacity is usually far larger than one user needs.



Point	TDM	FDM	WDM
Shares	time: each input gets a slot	frequency: each input gets a band	wavelength (colour of light)
Signal type	digital	analog	optical (analog in concept)
Medium	wire, fibre, radio	coax, radio (TV, FM)	optical fibre only
Separation	sync bits / guard time	guard bands	guard wavelength gap
Needs	tight synchronisation	filters, carriers	prisms / gratings
Problem	empty slots waste capacity	cross-talk, intermodulation	costly lasers
Example	T1/E1 lines, GSM slots	radio, cable TV, AMPS	DWDM backbone, SEA-ME-WE

Memory trick **T**DM = **T**urns in **T**ime; **F**DM = everyone talks at once on a different **F**loor (band); **W**DM = FDM done with **W**avelengths of light.

Q2 Differentiate among TDM, FDM and WDM. How does synchronous TDM work? Write about synchronous and statistical TDM.

✎ **Asked in:** Combined Bank SO(IT) 2023 · RAKUB ANSE 2023

Synchronous TDM: every input owns a **fixed slot** in every frame, whether it has data or not. Frame = one slot per input (+ framing bits). Output rate = n × input rate. Empty slot = wasted capacity.

Statistical TDM: slots are given **only to inputs with data**, so each slot carries an **address**. No fixed frame; link rate can be below n × input rate. Used by packet networks.

Exam trap "FDM is for analog, TDM for digital" is the standard line, but FDM can carry modulated digital data too (Wi-Fi OFDM). The 49th BCS MCQ asks TDM's **main disadvantage over FDM**: **complex synchronisation**.

Q3 What is multiplexing? Describe the different types of multiplexing, with examples, and explain time-division multiplexing in detail.

✎ **Asked in:** Combined Bank AME/AHE 2023 · SPCBL AME 2021 · SPCBL SAME 2022

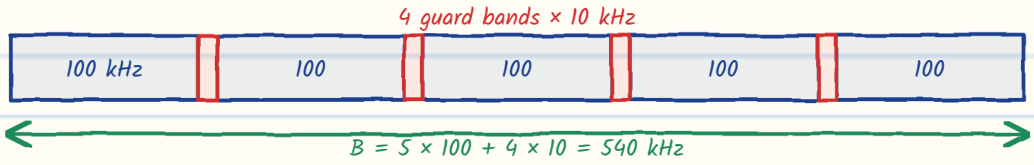
1. **FDM**: each signal modulates its own carrier; bands are separated by guard bands (radio, TV, 1G AMPS).
2. **WDM**: several light wavelengths on one fibre, combined by a prism / grating (DWDM: 40–160 λ).
3. **TDM**: synchronous (fixed slot per input) and statistical (slot on demand, with an address). Interleaving unit = bit, byte or block; a framing bit marks each frame.
4. **CDM / CDMA**: every user sends in the whole band at the same time, separated by orthogonal codes (3G).

2 FDM numericals: guard bands

bank IT 2026

Q4 Five channels, each with a 100-kHz bandwidth, are to be multiplexed together. What is the minimum bandwidth of the link if there is a need for a guard band of 10 kHz between the channels to prevent interference?

✎ **Asked in:** Combined Bank SO(IT) 2026



→ $B_{min} = N \cdot B + (N - 1) \cdot G = 500 + 40 =$ **540 kHz**

Memory trick Fence posts: N channels need only **N − 1** guard bands (none at the two edges).

Q5 Solve an FDM numerical: frequency allocation and required bandwidth (the figures are not legible on the surviving paper).
✎ **Asked in:** BSCL AE(CSE) 2026

The usual version: three 4-kHz voice channels share a link of 12 kHz, 20–32 kHz. Each channel modulates its own carrier and keeps the upper band: ch 1 → 20–24 kHz, ch 2 → 24–28 kHz, ch 3 → 28–32 kHz; the receiver separates them with band-pass filters and demodulates. No guard band here: if one is required, add $(N-1)\cdot G$ as in Q4.

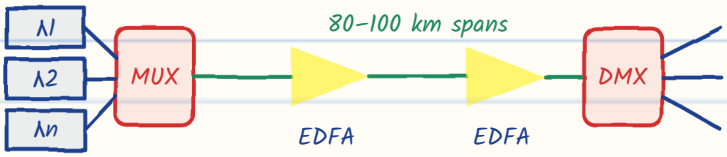
Q6 AMPS uses two bands: 824–849 MHz for sending and 869–894 MHz for receiving. Each user has a bandwidth of 30 kHz in each direction. How many people can use their cell phones simultaneously?
✎ **Asked in:** JBL ANA 2022

→ each band = 25 MHz; 25 MHz ÷ 30 kHz = 833.3 → AMPS uses **832 channels**; 42 are control channels → **790 users** at once (one channel in each band per user)

Exam trap Do not add the two bands (50 MHz ÷ 30 kHz): a call needs one 30-kHz channel **in each direction**, so the bands give the same count. If the paper ignores control channels, write 833 and say why.

Q7 Where are the low-loss transmission windows of silica optical fibre, and which window is the most popular? Draw a long-haul WDM transmission system.
✎ **Asked in:** BTCL AM(Technical) 2021

Window	Loss
850 nm	≈ 2–3 dB/km
1310 nm	≈ 0.35 dB/km
1550 nm	**≈ 0.2 dB/km**



1550 nm is the most popular: lowest loss, and **erbium-doped fibre amplifiers (EDFA)** amplify the whole 1530–1565 nm C-band at once, so dozens of DWDM channels cross thousands of km with only optical repeaters.

3 TDM frame numericals

bank & BTCL papers

Three rules solve every synchronous-TDM question ① **frame duration = duration of one input unit** (each input sends one unit per frame) · ② output slot = frame ÷ n · ③ output rate = n × input rate (+ framing bits × frame rate).

Q8 Four 1-kbps connections are multiplexed together. A unit is 1 bit. Find (a) the duration of 1 bit before multiplexing, (b) the transmission rate of the link, (c) the duration of a time slot and (d) the duration of a frame.
✎ **Asked in:** BB AME 2019 · also on an AHE/ANE suggestion set

→ (a) 1/1000 s = **1 ms** · (b) 4 × 1 kbps = **4 kbps** · (c) 1/4000 s = **250 µs** · (d) 4 slots × 250 µs = **1 ms**

Q9 The figure shows synchronous TDM with a data stream for each input and one data stream for the output. The unit of data is 1 bit. Find (a) the input bit duration, (b) the output bit duration, (c) the output bit rate and (d) the output frame rate.

✎ **Asked in:** BTCL AM(Technical) 2021 · JBL SO(ANE) 2020

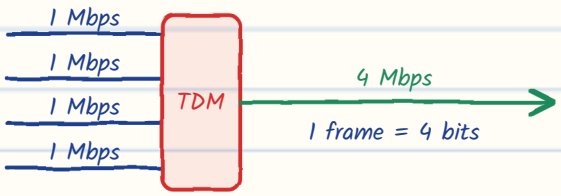


Figure (textbook version): four inputs at 1 Mbps.
(a) input bit = 1/1 Mbps = **1 µs**
(b) output bit = 1/4 Mbps = **0.25 µs**
(c) output rate = 4 × 1 Mbps = **4 Mbps**
(d) frame rate = input rate = **1,000,000 frames/s**

Q10 Two channels, one at 190 kbps and the other at 180 kbps, are multiplexed using pulse-stuffing TDM with no synchronisation bits. Find (a) the frame size in bits, (b) the frame rate, (c) the frame duration and (d) the data rate.

✎ **Asked in:** RAKUB ME 2021

→ pulse stuffing raises the 180-kbps channel to 190 kbps · (a) **2 bits** (1 per channel) · (b) **190,000 frames/s** · (c) 1/190,000 = **5.26 µs** · (d) 2 × 190 kbps = **380 kbps**

Q11 A TDMA system has 8 transmitter–receiver pairs. Each is sampled at 8 kHz, giving 16 bits per sample. Two synchronisation bits are added to each frame. Calculate the data rate of the TDMA link.

✎ **Asked in:** BTCL AM(Technical) 2021

→ frame = 8 × 16 + 2 = 130 bits; frames/s = 8000 → R = 130 × 8000 = **1.04 Mbps**

Q12 In a TDM system eight signals of 2 kHz each are sampled 25% above the Nyquist rate; the quantisation error must not exceed 1% of the signal amplitude. Find the bit rate (recalled wording).

✎ **Asked in:** NESCO AM(ICT) 2021

→ f_s = 1.25 × 2 × 2 kHz = 5 kHz · error ≤ Δ/2 = 1% of peak m_p, $\Delta = 2m_p/L$ → $L \geq 100$ → 128 levels, **$n = 7$** · R = 8 × 5000 × 7 = **280 kbps**

Q13 How long does it take to send a file of 500,000 bits from host A to host B over a circuit-switched network where all links are 1.536 Mbps, each link uses TDM with 32 slots, and it takes 800 ms to establish the end-to-end circuit?

✎ **Asked in:** Combined Bank AHE 2021 · also on the Petrobangla ANE suggestion sheet

→ one circuit = 1.536 Mbps ÷ 32 = 48 kbps · transfer = 500,000 ÷ 48,000 = 10.42 s · total = 10.42 + 0.8 = **11.22 s**

Exam trap A circuit gets **one slot**, i.e. 1/32 of the link, never the full 1.536 Mbps; and the set-up time is added once. Suggestion-sheet variant (16 slots, 1.5 Mbps, no set-up): 500,000 ÷ 93,750 = 5.33 s.

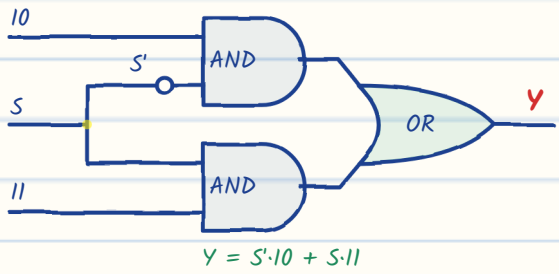
Drill A 2-Mbps TDM channel carries 500-bit frames. (a) Sampling rate per user = 2,000,000 ÷ 500 = **4000 samples/s**. (b) 8-bit samples + 4 sync bits per frame → (500 − 4) ÷ 8 = **62 users**. ✎ **Source:** Petrobangla ANE suggestion sheet

4 Multiplexer circuits

filed under multiplexing by the papers

These are digital-logic questions; the papers file them next to TDM. A **multiplexer (MUX)** is a data selector: 2^n inputs, n select lines, one output. A **demultiplexer (DEMUX)** sends one input to one of 2^n outputs.

Q14 Draw the logic circuit of a two-input multiplexer. Compare multiplexer and demultiplexer.
✎ **Asked in:** 45th BCS CSE (2024)



$Y = S'\cdot I0 + S\cdot I1$

Point	MUX	DEMUX
Lines	many → **one**	**one** → many
Selects	which input passes	which output gets it
Size	$2^n : 1$	$1 : 2^n$
Built from	AND-OR	decoder + data
Use	parallel → serial, TDM sender	serial → parallel, TDM receiver

Applications ✎ **Asked in:** Combined Bank Officer(IT) 2026: MUX picks one of several sensors for one ADC, or one of several registers onto a bus; DEMUX routes a bus to one of several devices (memory chip select, keyboard scanning).

Q15 Design a 4-to-1 multiplexer and explain its working.
✎ **Asked in:** 46th BCS ICT · DESCO SAE 2022 · 16th NTRCA CSE (2019)

$Y = S1'S0'I0 + S1'S0\ I1 + S1\ S0'I2 + S1\ S0\ I3$: four 3-input AND gates (each gets one data input and one select combination) feed a 4-input OR; the select code S1S0 = 00, 01, 10, 11 passes I0, I1, I2, I3.

Q16 Draw and write the output equation for an 8 × 1 multiplexer with an enable input E (E = L → output 0; E = H, S2S1S0 = LLL → D0 … HHH → D7).
✎ **Asked in:** Combined Bank SO 2020

→ $Y = E\cdot(S2'S1'S0'D0 + S2'S1'S0D1 + S2'S1S0'D2 + S2'S1S0D3 + S2S1'S0'D4 + S2S1'S0D5 + S2S1S0'D6 + S2S1S0D7)$

Q17 Draw a logic circuit for a 2-to-4 demultiplexer.
✎ **Asked in:** Rupali Bank ANE 2021

→ $Y0 = D\cdot S1'S0'$ · $Y1 = D\cdot S1'S0$ · $Y2 = D\cdot S1S0'$ · $Y3 = D\cdot S1S0$ (two inverters, four 3-input AND gates)

Q18 Design a three-input XOR, $f = w1 \oplus w2 \oplus w3$, using 2-to-1 multiplexers.
✎ **Asked in:** Dhaka WASA AME(Network) 2025

→ MUX 1: select w1, I0 = w2, I1 = w2' → $g = w1 \oplus w2$ · MUX 2: select w3, I0 = g, I1 = g' → $f = g \oplus w3$. Without inverters, a third MUX with I0 = 1, I1 = 0 makes g'.

MCQ	Answer
a. Parallel-to-serial converter: (a) decoder (b) encoder (c) demultiplexer (d) multiplexer ✎ **Asked in:** BUET IICT April 2021	**(d) multiplexer**
b. 2:1 MUX, output fed back to I0, data on I1, select S: (a) D flip-flop (b) D latch (c) half-adder (d) demultiplexer ✎ **Asked in:** BCIC AP 2025	**(b) D latch**
c. 8086 address lines multiplexed with data: (a) 8 (b) 32 (c) 64 (d) 16 ✎ **Asked in:** 49th BCS ICT (2025)	**(d) 16**

Exam trap (b) is a **latch**, not a flip-flop: while S = 1 the output follows D (level-sensitive); S = 0 holds it. (c) The 8086 multiplexes **AD0–AD15**: 16. "8" is the 8088.

5 The sampling theorem

38th, 46th BCS

Nyquist sampling theorem A signal band-limited to f_{max} is fully recovered from its samples if $f_s \geq 2f_{max}$. Nyquist rate = $2f_{max}$; Nyquist interval = $1/(2f_{max})$. Sampling slower causes **aliasing**: high frequencies fold back as false low ones.



Q19 Determine the Nyquist sampling rate and the Nyquist sampling interval for $x(t) = \sin(2100\pi t)$.
✎ **Asked in:** 38th BCS (2018)

1. Compare with $\sin(2\pi ft)$: $2\pi f = 2100\pi \rightarrow f_{max} = 1050$ Hz
2. Nyquist rate = $2 \times 1050 =$ **2100 samples/s**
3. Nyquist interval = $1/2100$ s ≈ **0.476 ms (476 µs)**

Exam trap The coefficient of t is $\omega = 2\pi f$, not f. Students write f = 2100 Hz and get 4200 samples/s. Some printed answers write "f = d/2" without a result: always finish with the number and the unit.

Q20 A message of bandwidth 20 kHz and amplitude 10 V is sampled at a rate 150% higher than the Nyquist rate and quantised uniformly; the bit rate is 1.6 Mbps. Find (i) bits per sample, (ii) the SQNR, (iii) the step size.

✎ **Asked in:** 46th BCS ICT

1. Nyquist rate = 40 kHz; "150% higher" = 40 × (1 + 1.5) = **100 kHz**
2. (i) n = 1.6 Mbps ÷ 100 kHz = **16 bits/sample**
3. (ii) sine input: SQNR = 1.76 + 6.02n = 1.76 + 96.32 = **98.1 dB**
4. (iii) $\Delta = 2A / 2^n$ = 20 V / 65536 ≈ **305 μV**

Exam trap "150% **higher** than" = 2.5 × Nyquist; "150% **of**" = 1.5 × Nyquist (60 kHz → 26.7 bits/sample, not a whole number, which shows it is the wrong reading).

MCQ	Answer
a. $g(t)$, band-limited to B Hz, has Nyquist samples 0 everywhere except 1 at $t = 0$. Then: (a) impulse at $t = 0$ (b) odd periodic (c) $g = 0$ for $\|t\| > 1/2B$ (d) $g = 0$ at $t = n/2B$, $n \neq 0$ ✎ **Asked in:** BUET M.Sc. April 2021	**(d) $g = 0$ at $t = n/2B$, $n \neq 0$**
b. $\sin(2\pi Bt)$ is band-limited to B Hz; $\sin^n(2\pi Bt)$ to: (a) 2^nB (b) B/n (c) nB (d) $2nB$ ✎ **Asked in:** BUET M.Sc. April 2021	**(c) nB**

(a) $g(t)$ is the **sinc** pulse, $\sin(2\pi Bt)/(2\pi Bt)$: zero at every other sampling instant, but not zero for all $\|t\| > 1/2B$.

(b) Multiplying in time = convolving spectra, so each power adds B: $\sin^2$ has $2B$, $\sin^n$ has nB.

Trick "Twice the **highest** frequency", never twice the bandwidth of a band-pass signal by default.

Drill 300–3400 Hz voice, PCM, error ≤ ±1% of full scale, range −1 V to +1 V: $\Delta/2 = 0.02$ V → $\Delta = 0.04$ V → $L = 2/0.04 = 50$ → **6 bits**; at the Nyquist rate 6.8 kHz → 40.8 kbps.

✎ **Source:** tail of an unidentified paper in a coaching compilation

6 Channel capacity: two formulas

13 exams

Nyquist (noiseless)

$R = 2B \log_2 L$

B = bandwidth, L = signal levels. More levels → more bits, until noise stops you.

Shannon (noisy)

$C = B \log_2(1 + SNR)$

SNR as a **ratio**: $SNR = 10^{dB/10}$. The ceiling no coding can beat.

Q21 A telephone line normally has a bandwidth of 3000 Hz (300 to 3300 Hz) assigned for data communication. The SNR is usually 3162. What will be the capacity of this channel?

✎ **Asked in:** Combined Bank AP 2023 · RPGCL AM(ICT) 2022

→ $C = 3000 \times \log_2(1 + 3162) = 3000 \times 11.63 =$ **34,881 bps ≈ 34.88 kbps** (3162 = 35 dB)

Exam trap One printed answer writes "SNR = 3162 dB": 3162 is the **power ratio** (= 35 dB). Putting 3162 dB into the formula is meaningless.

Q22 Find the channel capacity of a teleprinter channel with 300 Hz bandwidth and an SNR of 3 dB.

✎ **Asked in:** MRA AME 2022

→ $SNR = 10^{0.3} = 1.995 \rightarrow C = 300 \times \log_2(2.995) = 300 \times 1.583 =$ **474.8 bps** (one printed copy misprints 30 dB in the "given" line)

Q23 For a noisy channel, the signal-to-noise power ratio is 0.1 and the bandwidth is 2.4 GHz. What is the channel capacity in Mbps?

✎ **Asked in:** 46th BCS ICT

→ $C = 2.4 \times 10^9 \times \log_2(1.1) = 2.4 \times 10^9 \times 0.1375 =$ **330 Mbps** (SNR below 1 still gives capacity: spread-spectrum works this way)

MCQ	Answer
a. Gaussian channel, SNR = 15, B = 16 kHz: C (kbps) = (a) 32 (b) 64 (c) 18 (d) 128 ✎ **Asked in:** 49th BCS ICT (2025)	**(b) 64**
b. B = 1 MHz, SNR = 63: bit rate = (a) 4 Mbps (b) 6 Mbps (c) 8 Mbps (d) 10 Mbps ✎ **Asked in:** BPSC AME 2019	**(b) 6 Mbps**
c. B = 200 kHz, 4 signal levels, max data rate: (a) 400 kbps (b) 800 kbps (c) 1000 kbps (d) 1200 kbps ✎ **Asked in:** Combined Bank SO(IT) 2021	**(b) 800 kbps**

→ (a) $16k \times \log_2 16 = 64$ kbps · (b) $1M \times \log_2 64 = 6$ Mbps · (c) Nyquist: $2 \times 200k \times \log_2 4 = 800$ kbps

Q24 Using the Nyquist theorem, a channel can be sampled 12 million times per second; four-level signals give 2 bits per sample. What is the total data rate? (6-MHz noiseless channel.)

✎ **Asked in:** NESCO AM(ICT) 2021

→ $R = 2 \times 6\ \text{MHz} \times \log_2 4 =$ **24 Mbps**

Q25 Signal power 10 mW, noise power 1 μW: find SNR and SNR in dB.

✎ **Asked in:** MGMCL AM(ICT) 2022

→ $SNR = 10 \times 10^{-3} / 10^{-6} =$ **10,000** → $10 \log_{10}(10^4) =$ **40 dB**. Noiseless channel: SNR → ∞.

Drill 265 kbps over a noiseless 20-kHz channel: $265{,}000 = 2 \times 20{,}000 \times \log_2 L \rightarrow \log_2 L = 6.625 \rightarrow L \approx 98.7$ → use **128 levels** (7 bits) in practice. ✎ **Source:** worked example in a coaching compilation

Memory trick Shannon = **S**NR inside the log (**S**hannon **S**ays noise); Nyquist = **N**umber of levels inside. Use Shannon for the limit, Nyquist for the levels.

7 Bit rate, baud and decibels

short numericals

Key relations bit rate $N = r \times$ baud S, where $r = \log_2 L$ bits per signal element · $dB = 10 \log_{10}(P_2/P_1)$ · gains in dB **add** · −3 dB = half power · 10 dB = ×10.

Q26 An analog signal has a bit rate of 8000 bps and a baud rate of 1000 baud. How many data elements are carried by each signal element? How many signal elements are needed?

✎ **Asked in:** Sonali Bank Officer(IT) 2021

→ $r = 8000 / 1000 =$ **8 bits per signal element** → $L = 2^8 =$ **256 different signal elements**

Q27 A digitised TV picture uses 480 × 500 pixels, each with one of 32 intensity values; 30 pictures are sent per second. Find the source rate R.

✎ **Asked in:** BB AME 2023

→ $480 \times 500 = 240{,}000$ pixels × $\log_2 32 = 5$ bits × 30 /s = **36 Mbps**

Q28 How many 8-bit characters can be sent per second over a 9600-baud serial link in asynchronous mode with one start bit, eight data bits, two stop bits and one parity bit?

✎ **Asked in:** Combined Bank SO(IT) 2022

→ bits per character = 1 + 8 + 1 + 2 = 12 → 9600 ÷ 12 = **800 characters/s** (binary line: 1 baud = 1 bit/s)

Q29 (a) A signal is attenuated by 10 dB; what is its final power if it was originally 5 W? (b) A signal passes through three cascaded amplifiers of 4 dB gain each; total gain? (c) On a 5-kbps channel, how long does a 100,000-bit frame take to send?

✎ **Asked in:** BTCL AE(CSE) 2017

→ (a) −10 dB = ÷10 → **0.5 W** · (b) 4 + 4 + 4 = **12 dB** (power × 15.85) · (c) 100,000 ÷ 5000 = **20 s**

Q30 What is 3 dB? How can you turn a converter into an inverter?

✎ **Asked in:** BTRC AD(Technical) 2019

3 dB = a power ratio of 2 (10 log 2 = 3.01): −3 dB is half power (0.707 of voltage). The **3-dB bandwidth** is the band between the two half-power points of a filter or amplifier. **Converter → inverter:** a fully controlled (thyristor) rectifier works as a **line-commutated inverter** when its firing angle is above 90° and a DC source of the right polarity drives it: power then flows from DC to the AC side.

Q31 A link uses 16-QAM; the signal operates at 2400 baud. Calculate the data rate in bps.

✎ **Asked in:** RAKUB ANSE 2026

→ 16-QAM: $\log_2 16 = 4$ bits per baud → $4 \times 2400 =$ **9600 bps**

MCQ	Answer
a. Same bit rate: QPSK symbol time T_Q vs BPSK T_B: (a) $0.25T_B$ (b) $4T_B$ (c) $0.5T_B$ (d) $2T_B$ ✎ **Asked in:** 49th BCS ICT (2025)	**(d)** $2T_B$

QPSK carries 2 bits per symbol, so at the same bit rate each symbol lasts **twice** as long and needs half the bandwidth.

Drill 8-bit samples at 8 kHz: bit rate = $8 \times 8000 =$ **64 kbps**; on a binary (2-level) line the baud rate is also 64 kbaud. This is one PCM voice channel (DS0). ✎ **Source:** AHE/ANE suggestion set

Exam trap Baud counts **signal changes**, bps counts **bits**; they are equal only for 2-level signalling. dB values add only when they are gains or losses; dBm is an absolute power (0 dBm = 1 mW).

8 Modulation and AM

15 exams

Q32 What is modulation? Why is it necessary? What are modulation and demodulation? Write five uses.

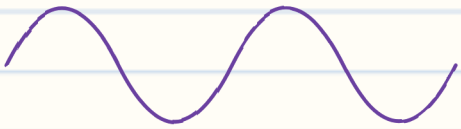
✎ **Asked in:** BPSC AE 2022 · BCSCL NE 2017

Definition Modulation varies a property (amplitude, frequency or phase) of a high-frequency **carrier** in step with the message. Demodulation recovers the message at the receiver.

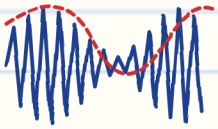
1. **Practical antenna size**: length ≈ λ/4; 3 kHz audio needs a 25-km antenna, 100 MHz needs 75 cm
2. **Multiplexing**: different carriers share one medium (FDM)
3. **Range**: high frequencies radiate and travel further
4. **Noise**: FM / digital schemes resist noise
5. **Matching**: shift the signal into the band the channel passes

Uses: ① AM/FM radio broadcast ② TV ③ mobile phones (GMSK, QAM) ④ modems / DSL ⑤ Wi-Fi, satellite and radar links.

Memory trick Why modulate? "**A**ntenna, **M**ultiplex, **R**ange, **N**oise" = **AMRN**.



message $m(t)$, f_m



AM: envelope = message



fc − fm · fc · fc + fm → B = 2fm

MCQ	Answer
a. Carrier 1 MHz, message 10 kHz, AM bandwidth: (a) 5 kHz (b) 10 kHz (c) 20 kHz (d) 2 MHz ✎ **Asked in:** 49th BCS ICT (2025)	**(c) 20 kHz**
b. Carrier 300 kHz, message 3 kHz, AM bandwidth: (a) 300 Hz (b) 3 kHz (c) 6 kHz (d) 600 Hz ✎ **Asked in:** DoICT ANE 2019	**(c) 6 kHz**
c. In amplitude modulation: (a) signal amplitude changes (b) signal frequency changes (c) carrier amplitude changes (d) carrier frequency changes ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(c) carrier amplitude changes**

Exam trap AM bandwidth = $2 \times f_m$, independent of the carrier: the 1-MHz carrier is a distractor. SSB halves it to f_m.

Q33 A DSB full-carrier AM wave is $v = 20(1 + 0.7 \sin 4500t) \sin 45000t$ volt. Find (a) the carrier frequency, (b) the modulating frequency, (c) the modulation index, (d) the lower and (e) the upper side-band frequency.
✎ **Asked in:** BTRC DAD 2021

→ $\omega_c = 45000$ rad/s → (a) $f_c = 45000/2\pi =$ **7162 Hz** · (b) $f_m = 4500/2\pi =$ **716.2 Hz** · (c) $m =$ **0.7** · (d) LSB = 7162 − 716 = **6446 Hz** · (e) USB = **7878 Hz**

Q34 Compare AM and FM in terms of bandwidth, noise performance and power efficiency.
✎ **Asked in:** 47th BCS ICT

Point	AM	FM
Varies	carrier amplitude	carrier frequency
Bandwidth	$2f_m$ (10 kHz broadcast)	$2(\Delta f + f_m)$ (200 kHz broadcast)
Noise	poor: noise adds to amplitude	good: limiter removes amplitude noise
Power	≥ 2/3 of power in the carrier	constant envelope, all power useful
Range / cost	long (sky wave), simple receiver	line of sight, complex receiver

9 FM numericals

BOF · 47th BCS

FM toolkit $\Delta f = k_f \times A_m$ · modulation index $\beta = \Delta f / f_m$ · **Carson's rule** $B = 2(\Delta f + f_m) = 2f_m(\beta + 1)$ · $\beta < 1$ narrowband ($B \approx 2f_m$).

Q35 A sinusoidal modulating wave of amplitude 5 V and frequency 2 kHz is applied to an FM generator with a frequency sensitivity of 40 Hz/V. Calculate the frequency deviation, the modulation index and the bandwidth.

✎ **Asked in:** BOF AP 2022

→ $\Delta f = 40 \times 5 =$ **200 Hz** · $\beta = 200 / 2000 =$ **0.1** (narrowband FM) · $B = 2(200 + 2000) =$ **4.4 kHz** ($\approx 2f_m = 4$ kHz)

Q36 A frequency-modulated voltage wave is $x(t) = 10 \cos(6.28 \times 10^8\, t + 5 \sin 1256t)$. (The paper is cut off after the equation; the usual parts are below.)

✎ **Asked in:** 47th BCS ICT

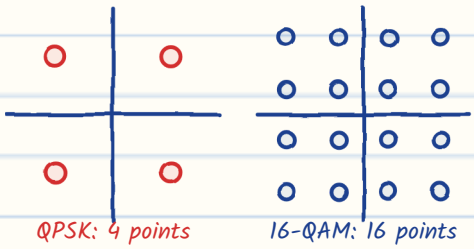
→ $f_c = 6.28 \times 10^8 / 2\pi \approx$ **100 MHz** · $f_m = 1256 / 6.28 =$ **200 Hz** · $\beta =$ **5** · $\Delta f = 5 \times 200 =$ **1 kHz** · $B = 2(1000 + 200) =$ **2.4 kHz** · power in 1 Ω = $10^2/2 =$ **50 W**

Exam trap In the equation the number in front of sin is **β**, not **Δf**; and $6.28 \approx 2\pi$, so divide it away. FM power never changes with β.

10 Digital modulation

49th BCS MCQs

Scheme	Varies	bits/symbol
ASK	amplitude	1
FSK / MSK / GMSK	frequency	1
BPSK	phase (0°, 180°)	1
QPSK	phase (4 states)	2
16-QAM	amplitude + phase	4
64-QAM	amplitude + phase	6

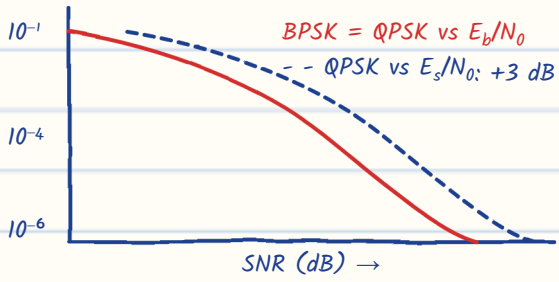


QPSK: 4 points 16-QAM: 16 points

MCQ	Answer
a. Most bandwidth-efficient: (a) BPSK (b) QPSK (c) FSK (d) 16-QAM ✎ **Asked in:** 49th BCS ICT (2025)	**(d) 16-QAM**
b. Modulation used in GSM: (a) MSK (b) FSK (c) ASK (d) GMSK ✎ **Asked in:** 49th BCS ICT (2025)	**(d) GMSK**

Q37 Draw the bit error rate (BER) vs signal-to-noise ratio curves of QPSK and BPSK.

✎ **Asked in:** NWPGCL AM(ICT) 2024



$P_b = Q(\sqrt{(2E_b/N_0)})$ for **both**. Values: 0 dB → 7.9×10^{-2}, 4 dB → 1.3×10^{-2}, 8 dB → 1.9×10^{-4}, 10 dB → 3.9×10^{-6}.

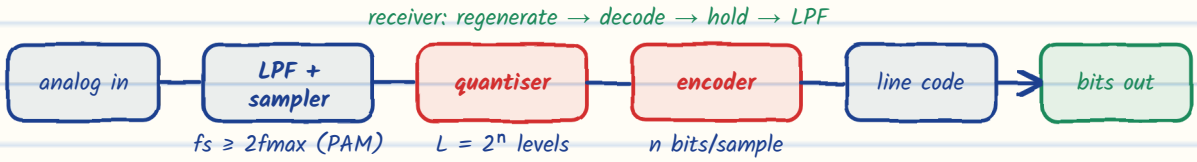
→ Against E_b/N_0 the curves **coincide**; against symbol SNR (E_s/N_0) QPSK sits **3 dB to the right**, because each symbol carries 2 bits. QPSK gives double the data rate in the same bandwidth for the same BER.

11 Pulse code modulation

38th BCS · BTCL

Q38 What is modulation? Describe the pulse code modulation (PCM) technique. State the advantages and disadvantages of PCM.

✎ **Asked in:** 38th BCS (2018) · also on the 46th & 47th BCS suggestion sheet



Steps. ① band-limit and **sample** at $f_s \geq 2f_{max}$ ② **quantise** each sample to one of $L = 2^n$ levels ③ **encode** each level as n bits ④ send as a line code. Bit rate = $n \cdot f_s$ (voice: 8×8000 = **64 kbps**).

Advantages	Disadvantages
noise immune; repeaters regenerate	needs far more bandwidth (64 k vs 4 kHz)
easy TDM, encryption, storage	quantisation noise always present
error detection / correction	complex coder; needs sync

Q39 A PCM system has a uniform quantiser with step size 2 V; the input is a sinusoid of amplitude 10 V. Calculate the signal-to-quantisation-noise ratio in dB and the number of bits per sample.

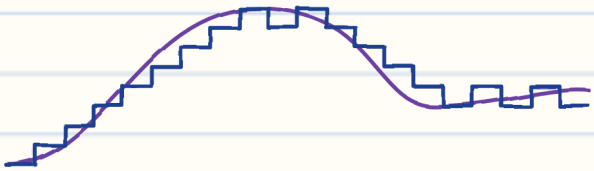
✎ **Asked in:** BTCL AM(Technical) 2021

1. Range = $2 \times 10 = 20$ V → $L = 20 / 2 = 10$ levels → $n = \lceil \log_2 10 \rceil =$ **4 bits**
2. Signal power = $A^2/2 = 50$; noise = $\Delta^2/12 = 4/12 = 0.333$
3. SQNR = $50 / 0.333 = 150$ → $10 \log 150 =$ **21.8 dB**

Exam trap 1.76 + 6.02n (= 25.8 dB for n = 4) assumes the levels fill the full range with $L = 2^n = 16$; here the step is fixed at 2 V, so only 10 levels are used: work from $A^2/2 \div \Delta^2/12$.

12 Delta modulation: one bit per sample

49th BCS · BUET



slope overload (step too small) granular noise (step too big)

DM sends 1 if the signal is above the staircase, 0 if below; the staircase moves one step Δ. **ADM** changes Δ with the slope (bigger on steep parts, smaller on flat parts), reducing **both** noises. **DPCM** sends the quantised difference with several bits.

MCQ	Answer
a. Higher slope-overload and granular noise: (a) DPCM (b) DM (c) ADM (d) PCM ✎ **Asked in:** 49th BCS ICT (2025)	**(b) DM**
b. True for delta modulation: (a) larger step ↓ granular noise (b) smaller step ↑ slope overload (c) smaller step ↑ granular noise (d) larger step ↑ slope overload ✎ **Asked in:** BUET M.Sc. April 2021	**(b) smaller step ↑ slope overload**

Memory trick **Small** step can't **climb** (slope overload); **big** step can't **sit still** (granular). ADM = adaptive step fixes both.

Hamming drills 4 data bits need r parity bits with $2^r \geq m + r + 1 \rightarrow r = 3$, code (7,4). Hamming **distance** = number of positions that differ: 10101 ⊕ 11110 = 01011 → **3**. ✎ **Source:** Petrobangla ANE suggestion sheet · worked example in a coaching compilation

13 Line coding

BPSC AE · SPCBL · BUET

Q40 What is line coding? Define the line-coding techniques and mention their categories.

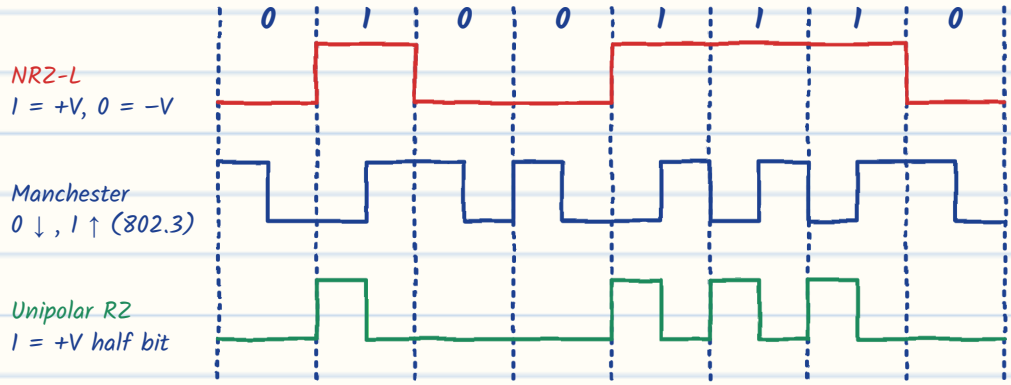
✎ **Asked in:** SPCBL AME 2021

Definition Line coding converts a sequence of bits into a digital signal (voltage levels) for the wire. A good code has no DC component, carries its own clock (self-synchronising), uses little bandwidth and detects some errors.

Category	Schemes	Note
Unipolar	NRZ, RZ (one level + zero)	DC component, rarely used
Polar	NRZ-L, NRZ-I, RZ, **Manchester**, **differential Manchester**	Manchester: Ethernet 10 Mbps; diff. Manchester: Token Ring
Bipolar	AMI, pseudoternary (B8ZS, HDB3 scrambling)	T1 / E1 lines
Multilevel	2B1Q, 8B6T, 4D-PAM5	ISDN, 100Base-T4, Gigabit
Multitransition	MLT-3	100Base-TX

Q41 We want to transmit the binary string 01001110. Show the resulting signal using (i) NRZ-L, (ii) Manchester, (iii) unipolar RZ.

✎ **Asked in:** BPSC AE 2022



Exam trap State your convention: some books draw NRZ-L with 0 = +V, and Manchester per G. E. Thomas is the inverse of IEEE 802.3. Any consistent convention earns full marks; an unstated one does not. Manchester always has a **mid-bit transition**.

Q42 A communication numerical on NRZ and RZ line coding (recalled wording).

✎ **Asked in:** Teletalk Executive(Technical) 2015

→ the usual ask is bandwidth for N bps: NRZ changes at most once per bit → $B_{min} \approx N/2$ (1 Mbps → 500 kHz); RZ and Manchester change twice per bit → $B_{min} \approx N$ (1 Mbps → 1 MHz), the price of self-clocking.

MCQ	Answer
a. 1110000100110101 in 4B/5B: max consecutive zeros in the output: (a) 1 (b) 4 (c) 2 (d) 3 ✎ **Asked in:** BUET M.Sc. April 2021	**(d) 3**
b. Read a differential Manchester waveform (figure not reproduced): (a) 10000111 (b) 01000100 (c) 01111000 (d) 10111011 ✎ **Asked in:** BUET M.Sc. April 2021	**see the rule below**

→ (a) 1110→11100, 0001→01001, 0011→10101, 0101→01011 → 11100 01001 10101 01011: longest run of 0s = **3** (4B/5B never allows more than 3). (b) Rule: a transition at the **start** of a bit = **0**, no transition = **1**; the mid-bit transition is only the clock.

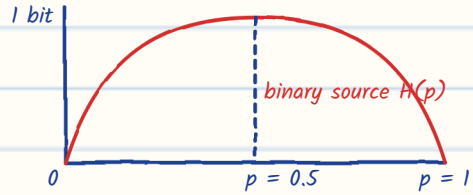
14 Entropy

47th, 49th BCS · DPDC

Q43 Explain the concept of entropy in information theory. How does it quantify the uncertainty of a random message source?

✎ **Asked in:** 47th BCS ICT

Definition The information in a symbol of probability p is $I = \log_2(1/p)$ bits: rare symbols carry more. Entropy is the **average information per symbol**: $H = -\Sigma\, p_i \log_2 p_i$ bits/symbol. It is largest ($\log_2 M$) when all M symbols are equally likely (most uncertain) and 0 when one symbol is certain. Information rate $R = r \times H$ (r = symbols per second).



certain event → H = 0
equally likely → H = $\log_2 M$ (max)

MCQ	Answer
a. The expression for entropy: (a) $\Sigma P_i \log_2 P_i$ (b) $-\Sigma \log_2 P_i$ (c) $-\Sigma P_i \log_2 P_i$ (d) $\Sigma P_i \log_2(P_i/(1+P_i))$ ✎ **Asked in:** 49th BCS ICT (2025)	**(c)** $-\Sigma P_i \log_2 P_i$

Q44 A discrete memoryless source emits A, B, C, D with probabilities 0.5, 0.25, 0.15, 0.10. (i) Calculate the entropy in bits per symbol. (ii) If one symbol is sent every millisecond, find the information rate.

✎ **Asked in:** 47th BCS ICT

Symbol	p	$\log_2(1/p)$	$p \cdot \log_2(1/p)$
A	0.50	1	0.5000
B	0.25	2	0.5000
C	0.15	2.737	0.4105
D	0.10	3.322	0.3322

→ (i) H = **1.743 bits/symbol** · (ii) r = 1000 symbols/s → R = 1000 × 1.743 = **1743 bps**

Q45 A source sends one of four symbols A, B, C, D as 2-bit pulses (00, 01, 10, 11); each bit lasts 5 ns. P(A) = 1/5, P(B) = 1/4, P(C) = 1/4, P(D) = 3/10. Calculate the average rate of information transmission.

✎ **Asked in:** DPDC AM(ICT) 2025

1. $H = 0.2 \log_2 5 + 2 \times 0.25 \log_2 4 + 0.3 \log_2(10/3) = 0.464 + 1.000 + 0.521 =$ **1.985 bits/symbol**
2. symbol time = 2 × 5 ns = 10 ns → $r = 10^8$ **symbols/s**
3. $R = r \times H = 10^8 \times 1.985 =$ **1.985×10^8 bps ≈ 198.5 Mbps**

Exam trap The raw bit rate is 200 Mbps (2 bits every 10 ns), but the **information** rate is lower (198.5 Mbps) because the symbols are not equally likely. Equal probabilities would give exactly 2 bits/symbol.

Memory trick Halves give whole bits: $p = ½ \rightarrow 1$ bit, $¼ \rightarrow 2$ bits, $\frac{1}{8} \rightarrow 3$ bits. Use this to check a table quickly.

15 Frequencies, bands and channels

BTRC · NTRCA · bank

Band	Range	Use
MF	300 kHz–3 MHz	AM radio
HF	3–30 MHz	short wave
VHF	**30–300 MHz**	FM, TV
UHF	300 MHz–3 GHz	mobile, Wi-Fi
SHF	3–30 GHz	satellite, microwave
EHF	30–300 GHz	mmWave, 5G

MCQ	Answer
a. VHF range: (a) 3–30 MHz (b) 30–300 MHz (c) 300 MHz–3 GHz (d) 3–30 GHz ✎ **Asked in:** 49th BCS ICT (2025)	**(b) 30–300 MHz**

Trick Every band is a ×10 step that starts with **3**: 3 k, 30 k, 300 k, 3 M, 30 M (VHF starts here), 300 M …

Q46 What is the difference between audio frequency and radio frequency?
✎ **Asked in:** BTRC DAD 2021 · BTRC DAD(Technical) 2019

Audio frequency (AF): 20 Hz–20 kHz, what the ear hears; travels as sound (pressure waves) and cannot radiate efficiently from an antenna. **Radio frequency (RF)**: about 3 kHz–300 GHz electromagnetic waves that radiate from an antenna and carry AF as modulation. AF = message, RF = carrier.

Q47 Dhaka radio broadcasts on medium wave at "630 Hz". If radio waves travel at 3×10^8 m/s, what is the wavelength?
✎ **Asked in:** BTRC AD(Technical) 2019 · BTRC SAD 2021

→ $\lambda = c/f = 3 \times 10^8 / 630 =$ **476,190 m** as printed; the station's real frequency is **630 kHz** → $\lambda =$ **476.2 m** (medium wave = hundreds of metres)

MCQ	Answer
b. Period 10 ms: frequency = (a) 10 Hz (b) 20 Hz (c) 50 Hz (d) 100 Hz ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(d) 100 Hz**
c. Bursts of 697 Hz and 1477 Hz: key pressed = (a) 1 (b) 2 (c) 3 (d) 4 ✎ **Asked in:** Combined Bank SO(IT) 2021	**(c) 3**

(c) **DTMF**: each key = one row tone + one column tone. Rows 697 / 770 / 852 / 941 Hz; columns 1209 / 1336 / **1477** Hz. 697 = top row (1 2 3), 1477 = third column → **3**.

Q48 Differentiate low-pass and band-pass channels with examples.
✎ **Asked in:** 15th NTRCA ICT

Low-pass: passes 0 to f_1 (starts at zero) → baseband digital signals go straight on it (a LAN cable). **Band-pass**: passes f_1 to f_2, not starting at zero → the signal must be **modulated** to fit (a radio or telephone voice channel, 300–3400 Hz). A low-pass channel is usually one wide medium used by one pair of devices; a band-pass channel is a slice of a shared medium.

Q49 What is the main benefit of a broadband transmission system compared with baseband?
✎ **Asked in:** Rupali Bank ANA 2023

Baseband puts one digital signal on the whole bandwidth of the cable (Ethernet). **Broadband** modulates signals onto separate carriers, so many channels travel at once (FDM: cable TV, DSL, cable Internet) and over longer distances with amplifiers.

Q50 What is channel hardening, and what does it imply for frequency allocation (in OFDMA, for example)?
✎ **Asked in:** BTRC DAD(Technical) 2019

In **massive MIMO**, with many base-station antennas the fading on the individual paths averages out, so the effective channel gain of each user becomes almost **constant** (deterministic) over time and frequency. Then OFDMA need not chase the best sub-carriers per user (frequency-selective scheduling); every user can get the whole band, with fewer pilots and simpler power control.

Exam trap $\lambda = c/f$: kHz → hundreds of metres, MHz → metres, GHz → centimetres. A wavelength of 476 km for a broadcast station signals a unit slip.

16 Simplex, half-duplex, full-duplex

NTRCA · BUET · 49th BCS

Q51 Write the differences between simplex, half-duplex and duplex communication with examples.
✎ **Asked in:** 15th NTRCA ICT

Mode	Direction	Capacity	Example
Simplex	one way only	whole channel one way	keyboard → CPU, TV / radio broadcast
Half-duplex	both ways, **one at a time**	whole channel, taking turns	walkie-talkie, CB radio
Full-duplex	both ways **at once**	split / two paths	telephone, mobile call

MCQ	Answer
a. Mode used during a phone conversation: (a) simplex (b) full-duplex (c) half-duplex (d) multiplex ✎ **Asked in:** 49th BCS CSE (2025)	**(b) full-duplex**
b. New phone: a user cannot talk while the other talks, but either side may start: (a) simplex (b) half-duplex (c) full-duplex (d) none ✎ **Asked in:** BUET M.Sc. April 2021	**(b) half-duplex**
c. Which mode uses serial communication? (a) duplex (b) half duplex (c) simplex (d) all ✎ **Asked in:** BCC AP 2021	**(d) all**

Exam trap (b) "one-way at a time, from either side" is **half-duplex**, not simplex: simplex can never reverse. (c) Serial / parallel describes how bits travel; direction is a separate property, so all three modes can be serial.

17 Synchronous vs asynchronous transmission

RAKUB · NTRCA · WZPDCL

QS2 Write the differences between synchronous and asynchronous transmission, with advantages and disadvantages. What is the problem of synchronous transmission, and how does asynchronous transmission solve it?

✎ **Asked in:** RAKUB ANSE 2023 · 15th NTRCA Computer Operation (2019) · WZPDCL AE(CSE) 2019



synchronous: one long block of bytes, framed by sync characters, shared clock

Point	Asynchronous	Synchronous
Unit	one character, start + stop bits	a block / frame of many bytes
Clock	re-synchronised by each start bit	shared or recovered clock
Gaps	random idle time between bytes	none: continuous stream
Overhead / speed	≥ 20% (2 of 10 bits); slow	small; fast
Cost / use	cheap; keyboard, RS-232, UART	needs buffers; LAN, SONET, HDLC

The problem with synchronous mode: sender and receiver must keep their clocks locked for the whole block; a small drift over thousands of bits misplaces every later bit, and the receiver must buffer the block. **The asynchronous fix:** each character brings its own start bit, so the receiver re-locks every 10 bits; drift never accumulates, at the cost of the start/stop overhead.

Q53 Name and define the five components of a data communication system, with a diagram.

✎ **Asked in:** BPSC AP 2023



Message = the data (text, voice, video); **sender / receiver** = the devices; **medium** = the physical path (cable, fibre, radio); **protocol** = agreed rules (format, timing, error handling). Without a protocol two connected devices exchange signals but no information.

Q54 What is data transmission speed? Discuss its types.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

The number of bits sent per second (bps). **Narrow band**: up to about 300 bps (telegraph) · **Voice band**: up to about 9600 bps–56 kbps (telephone line, modem) · **Broadband**: 1 Mbps and above (fibre, DSL, coax, microwave, satellite).

18 Transmission media

bank & NTRCA

Q55 What types of transmission media are used for data communication? Explain their advantages and disadvantages. What is attenuation? Distinguish twisted pair, coaxial cable and optical fibre in a table.

✎ **Asked in:** Combined Bank SO(IT) 2024 · Rupali Bank ANA 2023 · 15th NTRCA ICT

Guided (wired): twisted pair, coaxial, optical fibre. **Unguided** (wireless): radio waves, microwave, infrared, satellite. Attenuation = loss of signal power with distance, in dB (or dB/km); amplifiers / repeaters restore it.

Point	Twisted pair	Coaxial	Optical fibre
Carries	electrical signal	electrical signal	**light**
Bandwidth	low (≤ 10 Gbps short)	medium	highest
Attenuation	high: repeater ~100 m	medium	lowest: 0.2 dB/km
EMI / tapping	affected (twist reduces)	shield helps	immune, hard to tap
Cost / install	cheapest, easy	moderate	costly, skilled splicing
Use	LAN (Cat-6), telephone	cable TV, CCTV	backbone, submarine

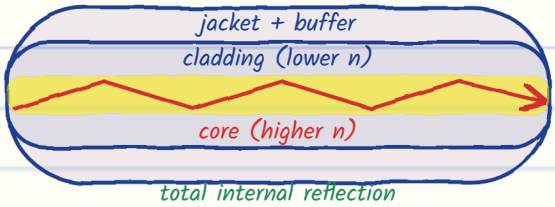
MCQ	Answer
a. A guided medium: (a) radio wave (b) microwave (c) coaxial cable (d) infrared ✎ **Asked in:** 49th BCS CSE (2025)	**(c) coaxial cable**
b. Cable for an analog CCTV camera: (a) UTP (b) coaxial (RG-59) (c) fibre (d) HDMI ✎ **Asked in:** Combined Bank Officer(IT) 2026	**(b) coaxial (RG-59)**
c. Needs line of sight: (a) microwave (b) fibre (c) twisted pair (d) coaxial ✎ **Asked in:** BPSC AP 2022	**(a) microwave**
d. A disadvantage of optical fibre: (a) resists corrosion (b) immune to tapping (c) unidirectional light (d) none of these ✎ **Asked in:** Combined Bank AHE/AE(IT) 2021	**(c) unidirectional light**

Exam trap (b) The options of this paper are not preserved; the answer it wants is **coaxial**. (d) The printed key says "none of these", but (a) and (b) are advantages; one-way light (two fibres for two-way traffic) is the textbook **disadvantage**, so answer (c).

19 Optical fibre

NTRCA · BPDB · BTCL · IBBL

Q56 Explain the structure and characteristics of fibre-optic cable, its advantages and disadvantages, and how data travels through it.
✎ **Asked in:** 17th NTRCA ICT (2023) · 13th NTRCA CSE (2016)



Structure: glass **core** (9 µm single-mode, 50/62.5 µm multimode), **cladding** with a lower refractive index, buffer coating, strength members (Kevlar), outer jacket. **How data travels:** a laser / LED sends light pulses; light hitting the core–cladding boundary above the critical angle is totally reflected back, so it stays in the core.

Advantages: huge bandwidth, very low loss, immune to EMI, light and thin, no sparks, hard to tap. **Disadvantages:** costly installation and splicing, fragile, light travels one way, needs O/E conversion.

Q57 What are the main requirements of optical fibre splicing?
✎ **Asked in:** BPDB AE(CSE) 2021

① exact **core alignment** (lateral, angular, gap) ② clean, flat, perpendicular **cleave** ③ **low insertion loss** (fusion ≈ 0.05–0.1 dB) and low back-reflection ④ mechanical strength: protective sleeve / splice tray ⑤ dust-free, well-matched fibres. Fusion splicing (arc-welded) beats mechanical splicing on loss.

Q58 Single-mode fibre 0.35 dB/km, splitter loss 14 dB, two splices of 0.1 dB, two connectors of 0.75 dB, transmitter 5 dBm, receiver sensitivity −14 dBm. Find the maximum OLT–ONU fibre length.

✎ **Asked in:** IBBL SO(Network) 2025

→ budget = 5 − (−14) = 19 dB · fixed losses = 14 + 0.2 + 1.5 = 15.7 dB · left for fibre = 3.3 dB → L = 3.3 / 0.35 = **9.43 km**

Q59 A 1550-nm link is 50 km long without repeaters; the single-mode fibre loses 0.2 dB/km and is joined every 2 km with connectors of 0.5 dB each. Find the minimum average launch power for 10 μW at the receiver.

✎ **Asked in:** BTCL AM(Technical) 2021

→ fibre 50 × 0.2 = 10 dB · joints every 2 km = 24 × 0.5 = 12 dB · total 22 dB · 10 μW = −20 dBm → launch = −20 + 22 = **+2 dBm ≈ 1.58 mW**

Exam trap 25 segments of 2 km have **24** joints between them (the end connectors belong to the transmitter and receiver). Work in dB and dBm, then convert once at the end.

20 Satellite, submarine cable, mmWave

RAKUB · BSCPLC · Petrobangla

Q60 What are optical cable and satellite transmission? Differentiate them. Name Bangladesh's submarine cables. Why is fibre used under the sea instead of satellite, and why do submarine cables get damaged?

✎ **Asked in:** RAKUB ANSE 2023 · BSCPLC AM(Engineering) 2024

Fibre: light in glass, terabits, ≈ 5 μs/km delay, cheap per bit, repairs need a ship. **Satellite**: radio via a GEO satellite at 35,786 km, ≈ 240–270 ms one-way delay, limited bandwidth, weather fade, but covers remote areas, ships and planes. Hence submarine fibre carries the bulk of international traffic. Bangladesh: **SEA-ME-WE 4** (Cox's Bazar), **SEA-ME-WE 5** (Kuakata), **SEA-ME-WE 6** (Cox's Bazar). Damage: ship anchors and fishing trawls (most faults), earthquakes and undersea landslides, abrasion on rocks, shark bites in old cables.

Q61 What is unguided media? Write the features of satellite communication.

✎ **Asked in:** 18th NTRCA School-2 (2024)

Unguided media carry electromagnetic waves through air / space without a conductor (radio, microwave, infrared, satellite). **Satellite features:** wide coverage (3 GEO satellites cover the globe), distance-insensitive cost, broadcast to many receivers, uplink/downlink on different frequencies (C, Ku, Ka band), long delay, rain fade, very high launch cost.

MCQ	Answer
a. VSAT stands for: (a) Virtual Small Aperture Satellite (b) Very Small Aperture Terminal (c) Very Small Application Terminal (d) Vertical Satellite ✎ **Asked in:** BDCCL AM(Transmission) 2022	**(b) Very Small Aperture Terminal**
b. Online banking from a plane or a ship needs: (a) fibre (b) satellite (c) microwave (d) coaxial ✎ **Asked in:** BB AD(General) 2022	**(b) satellite**
c. Bluetooth uses: (a) infrared (b) radio waves (2.4 GHz) (c) laser (d) microwave dish ✎ **Asked in:** BB AD(General) 2022	**(b) radio waves (2.4 GHz)**

Q62 What is the purpose and primary application of millimetre waves (mmWave) in IoT ecosystems?
✎ **Asked in:** Petrobangla AM(CSE) 2024

mmWave = 30–300 GHz (5G uses 24–100 GHz). **Purpose:** very wide channels → multi-Gbps rates and about 1 ms latency for dense device clusters; small antennas allow beam-forming arrays. **Applications:** 5G fixed wireless and backhaul, industrial IoT / smart factories, AR/VR, connected vehicles (V2X), high-resolution radar sensing. **Limit:** short range, blocked by walls and rain, so it needs dense small cells.

➜ Satellite orbits at a glance

Orbit	Height	One-way delay	Examples
LEO	≈ 500–2000 km	≈ 5–20 ms	Starlink, Iridium, imaging satellites
MEO	≈ 2000–35,786 km	≈ 50–150 ms	GPS (≈ 20,200 km), O3b
GEO	**35,786 km**	**≈ 240–270 ms**	TV broadcast, VSAT, Bangabandhu Satellite-1 (119.1° E)

Memory trick **GEO** = **G**oes round **E**xactly **O**nce a day: it stays fixed over one point, so dishes never move, but every hop costs about a quarter of a second.

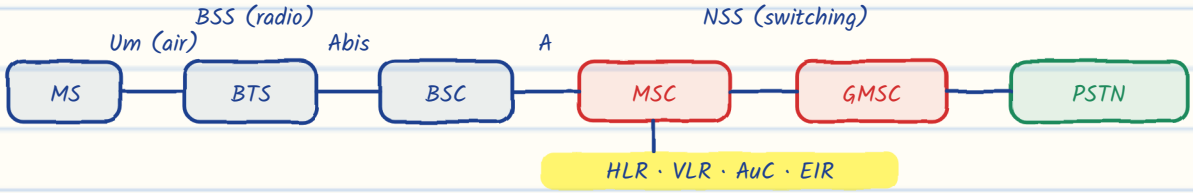
Viva rapid-fire Satellite vs submarine cable → delay, capacity, cost (Q60) · SEA-ME-WE = South-East Asia–Middle East–Western Europe · landing stations: **SMW-4 Cox's Bazar, SMW-5 Kuakata (Patuakhali)** · fibre vs Ethernet copper: fibre for long / noisy runs, copper for short desk links; both lose data only through errors, not by design · light stays in fibre by total internal reflection (core n > cladding n) · first fibre telephone links: 1977, USA. ✎ **Source:** bank IT viva (candidates' accounts) · Rupali Bank ANE viva

21 Mobile systems: GSM, GPRS, LTE

BTRC · Teletalk · bank

Q63 Draw the GSM architecture. Write the full meaning of GSM and CDMA. What are the functions of Node B and GGSN?

✎ **Asked in:** Teletalk Executive(Technical) 2015 · Teletalk Executive 2019 · Combined Bank SO(IT) 2020



GSM = Global System for Mobile communications; **CDMA** = Code Division Multiple Access. **Node B**: the 3G (UMTS) base station: WCDMA radio, modulation, power control, controlled by the RNC. **GGSN** (Gateway GPRS Support Node): gateway between the mobile packet core and external networks (Internet); assigns IP addresses, routes packets to the right SGSN, applies policy, firewall and charging.

Q64 What is LTE (full meaning)? Name its advanced technology. Explain LTE and GSM and the differences between them; describe LTE radio technology; list the LTE network elements.

✎ **Asked in:** BTRC AD 2021 · BTRC DAD 2021 · BTRC AD(Technical) 2019 · Combined Bank Officer(IT) 2020

LTE = Long Term Evolution (4G, 3GPP Release 8); advanced forms **LTE-Advanced** (Rel. 10: carrier aggregation, 8×8 MIMO) and LTE-Advanced Pro. Radio: OFDMA downlink, SC-FDMA uplink, MIMO, 1.4–20 MHz channels, all-IP. **Elements:** UE, **eNodeB** (E-UTRAN), and the EPC: **MME** (signalling, mobility), **S-GW**, **P-GW** (Internet gateway, IP address), **HSS** (subscriber database), PCRF (policy).

Point	GSM (2G)	LTE (4G)
Access	TDMA + FDMA	OFDMA / SC-FDMA
Switching	circuit (voice)	packet, all-IP (VoLTE)
Channel	200 kHz	up to 20 MHz
Data rate	9.6 kbps (GPRS ~ 114)	100+ Mbps
Base station	BTS + BSC	eNodeB only

Q65 What is GPRS? Write its features and advantages.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

GPRS (General Packet Radio Service, "2.5G") adds packet switching to GSM: SGSN + GGSN nodes, up to 171 kbps theoretically (≈ 40–114 kbps real). Features: always-on, billed by data volume, timeslots shared by many users, IP-based (WAP, email, MMS). Advantages: efficient use of radio, instant access, the base for EDGE and 3G cores.

Q66 What is the function of TCSM? What equipment is needed for VoIP? What is a call drop and what causes it? What are GPS, VoIP, CDMA and GSM, and why are they used?

✎ **Asked in:** BTRC DAD 2021 · BTRC DAD(Technical) 2019

1. **TCSM** (transcoder and sub-multiplexer): converts the 13-kbps GSM speech (16 kbps with overhead) to 64-kbps PCM for the MSC, and packs four 16-kbps channels into one 64-kbps timeslot, saving transmission lines.
2. **VoIP equipment:** IP phone or softphone / ATA adapter, broadband IP network with QoS, SIP server / IP-PBX (softswitch), VoIP gateway to the PSTN, codecs (G.711, G.729), router / firewall.
3. **Call drop** = a call cut off before either user ends it. Causes: weak coverage, handover failure, interference, network congestion, faulty BTS / backhaul, high speed of the user, handset fault.
4. **GPS**: satellite positioning (navigation, timing) · **VoIP**: voice over IP networks, cheap calls · **CDMA**: users share the band by codes · **GSM**: 2G digital cellular standard for voice and SMS.

Q67 Discuss the advantages and disadvantages of TDMA and CDMA.

✎ **Asked in:** 46th BCS ICT

	TDMA	CDMA
Advantages	simple, cheap handsets, battery saving (radio off in other slots), easy hand-off	more users per MHz, soft hand-off, resists interference and tapping, frequency reuse 1
Disadvantages	needs strict sync and guard time, fixed slots wasted, multipath ISI	needs tight power control (near–far problem), complex receivers, capacity falls as load rises

Q68 Identify the differences between 1G, 2G, 3G and 5G networks. Write the disadvantages of 5G in terms of initial cost and battery drainage.

✎ **Asked in:** Fisheries & Livestock Instructor(CS) 2023 · Rupali Bank ANE 2021

Gen	Technology	Signal / access	Speed	Service
1G	AMPS	analog, FDMA	voice only	voice
2G	GSM, IS-95	digital, TDMA / CDMA	9.6–64 kbps	voice, SMS
3G	UMTS / WCDMA	WCDMA	≈ 2 Mbps+	video call, mobile Internet
5G	NR	OFDM, mmWave, massive MIMO	up to 20 Gbps, ≈ 1 ms	eMBB, IoT, URLLC

5G disadvantages: initial cost: dense small cells, fibre backhaul and new spectrum make roll-out very expensive; **battery drainage**: wide-band radios and many antennas use more power and heat, so phones drain faster, especially while searching between 4G and 5G.

Q69 Write the differences between a digital telephone and a mobile phone. What is a modem? Write its uses.

✎ **Asked in:** 15th NTRCA Computer Operation (2019)

Digital (fixed) telephone: wired to the exchange, fixed location, digital switching (ISDN / PCM), very stable. **Mobile phone**: wireless cellular link to a base station, moves with hand-over, needs a SIM and battery, adds SMS and data. **Modem** (modulator–demodulator) converts digital data to an analog signal for the line and back. Uses: dial-up / DSL / cable Internet, fax, GSM modems for SMS alerts, POS terminals, remote monitoring.

Q70 Explain the parity method for error detection. Write the bit strings of "Delta" using ASCII.

✎ **Asked in:** BB AME 2023

The sender adds **one parity bit** so the count of 1s is even (even parity) or odd (odd parity); the receiver recounts. It detects any **odd number** of flipped bits, never an even number, and cannot correct.

Char	ASCII	7-bit	even parity (MSB)
D	68	1000100	01000100
e	101	1100101	01100101
l	108	1101100	01101100
t	116	1110100	01110100
a	97	1100001	**11100001**

MCQ	Answer
a. Two-tier client/server: application and UI run on the: (a) server side (b) host side (c) modulation side (d) none (client side) ✎ **Asked in:** Combined Bank ADA 2021	**(d) none (client side)**
b. Asymmetric encryption: the key that encrypts is the: (a) private key (b) encryption key (c) public key (d) modulation key ✎ **Asked in:** Combined Bank ADA 2021	**(c) public key**
c. Switching delayed because data is stored in and read from RAM: (a) space-division (b) time-division (c) virtual (d) packet ✎ **Asked in:** Combined Bank SO(IT) 2021	**(b) time-division**

Viva rapid-fire Name some cables → twisted pair (UTP/STP), coaxial, optical fibre · what is a medium → the physical path between sender and receiver; guided or unguided · SIM → Subscriber Identity Module: a smart card holding the IMSI and the key Ki that the network uses to authenticate the subscriber · a digital signal → discrete levels (draw a square wave) · polarisation → the direction of the electric field of a wave (vertical, horizontal, circular); antennas must match it · Ethernet sends digital data as voltage levels (line-coded), not an analog carrier. ✎ **Source:** Combined Bank Officer (IT) 2026 viva · BPSC viva 2023 · bank IT viva · Rupali Bank ANE viva

☾ Night before the exam

one page, every formula

Formulas Nyquist rate = $2f_{max}$ ($\omega = 2\pi f$!)
Nyquist $R = 2B \log_2 L$ · Shannon $C = B \log_2(1+SNR)$
$SNR = 10^{dB/10}$ · 3 dB = ×2 · 10 dB = ×10
bps = baud × $\log_2 L$ · QAM-16 = 4 bits/baud
FDM $B = NB + (N-1)G$ · TDM out = n × in
AM $B = 2f_m$ · FM $B = 2(\Delta f + f_m)$, $\beta = \Delta f/f_m$
PCM $R = n \cdot f_s$ · SQNR = 1.76 + 6.02n dB
$H = -\Sigma p \log_2 p$ · $R = r \cdot H$ · $\lambda = c/f$

One-liners TDM = time slots, digital, needs sync · FDM = bands, analog, guard bands · WDM = light, fibre
GSM → GMSK · most efficient → 16-QAM · DM noise → slope overload + granular
VHF 30–300 MHz · DTMF 697 + 1477 = "3"
Manchester = mid-bit transition · 4B/5B ≤ 3 zeros
fibre windows 850 / 1310 / **1550** nm · SMW-4, 5, 6
LTE: OFDMA ↓ SC-FDMA ↑ · GGSN = Internet gateway

➜ Which formula for which wording

The question says …	Use	See
"guard band", "minimum bandwidth of the link"	$NB + (N-1)G$	Q4
"unit is 1 bit", "frame duration / slot"	frame = input unit time; slot = frame ÷ n	Q8–Q11
"SNR", "noisy channel", "capacity"	Shannon, SNR as a ratio	Q21–Q23
"noiseless", "signal levels"	Nyquist $2B \log_2 L$	Q24
"sin(ωt)", "Nyquist rate / interval"	$f = \omega/2\pi$, then 2f	Q19
"step size", "SQNR"	$(A^2/2) \div (\Delta^2/12)$	Q39
"probabilities", "information rate"	H, then $R = r \cdot H$	Q44, Q45
"dB/km", "sensitivity", "launch power"	power budget in dB	Q58, Q59

All the tricks Turns-in-Time / Floors / Wavelengths · fence posts (N − 1 guard bands) · Shannon Says noise · AMRN (why modulate) · small step can't climb · halves give whole bits · bands start with 3.

Top traps ω vs f in $\sin(2100\pi t)$ · SNR ratio vs dB · "150% higher" = 2.5× · AM bandwidth ignores the carrier · a circuit gets one TDM slot · latch vs flip-flop · 24 joints in 50 km at 2-km spacing.

✓ Practice MCQs (asked ones first)

try first, then check

1. Main disadvantage of TDM over FDM: ✎ 49th BCS ICT (2025)

(a) higher bandwidth (b) noise immunity
(c) complex sync (d) limited frequency

2. Which multiplexing transmits digital signals? ✎ BPSC AME 2019

(a) WDM (b) FDM
(c) TDM (d) all of these

3. SNR = 63, B = 1 MHz: bit rate = ✎ BPSC AME 2019

(a) 4 Mbps (b) 6 Mbps
(c) 8 Mbps (d) 10 Mbps

4. Combines signals by assigning time slots: ✎ 49th BCS CSE (2025)

(a) TDM (b) FDM
(c) WDM (d) SDM

5. AM: carrier 300 kHz, message 3 kHz, bandwidth: ✎ DoICT ANE 2019

(a) 300 Hz (b) 3 kHz
(c) 600 Hz (d) 6 kHz

6. VSAT: ✎ BDCCL AM(Transmission) 2022

(a) Very Small Aperture Terminal (b) Virtual Small Aperture Satellite
(c) Very Small Application Terminal (d) Vertical Satellite

7. Codec digitising analog signals uses:

(a) demodulation (b) PCM
(c) CDMA (d) NRZ

8. LTE means:

(a) Long Term Evaluation (b) Long Time Evaluation
(c) Long Terminal Evolution (d) Long Term Evolution

9. Nyquist rate of $\sin(2100\pi t)$:

(a) 1050 Hz (b) 4200 Hz
(c) 2100 Hz (d) 6600 Hz

10. Bits per baud in 64-QAM:

(a) 6 (b) 4
(c) 8 (d) 64

Answer key → 1-c · 2-c · 3-b · 4-a · 5-d · 6-a · 7-b · 8-d · 9-c · 10-a

Written question What is multiplexing? Write the differences between TDM and FDM, and find the minimum link bandwidth for five 100-kHz channels with 10-kHz guard bands.

10 marks

✎ **Asked in:** 45th BCS CSE (2024) · BPSC AP 2021 · Combined Bank SO(IT) 2026

1. Definition: several signals share one link; MUX at the sender, DEMUX at the receiver; why: link capacity ≫ one signal — 2
2. Diagram: MUX → one link → DEMUX, TDM slots vs FDM bands — 2
3. Table (at least 5 points): shares time vs frequency, digital vs analog, sync bits vs guard bands, needs synchronisation vs filters, examples E1 vs radio — 4
4. $B = 5 \times 100 + (5 - 1) \times 10 =$ **540 kHz**, with the four guard bands drawn — 2

→ Before the exam ✓

- ✓ TDM vs FDM vs WDM table
- ✓ TDM frame: duration, slot, rate
- ✓ Shannon with SNR in dB
- ✓ Nyquist rate from ω
- ✓ PCM blocks + pros/cons
- ✓ AM / FM bandwidth, β